Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
December-2024

Max. Marks: 100 **Duration: 3 Hrs**

Class: TE **Semester: V**

Course Code: CS394 **Programme: CE**

Name of the Course: Distributed Computing

Instructions:

1. **All questions are compulsory.**
2. **Draw neat diagrams and write equations wherever necessary.**
3. **Assume suitable data if necessary**

Course Outcomes: At the end of the course students will be able to:

1. **Understand the Principles and desired Properties of distributed Systems.**
2. **Apply the Various Communication techniques for distributed communication.**
3. **Apply the concepts of process, naming, consistency, replication, and fault tolerance in a distributed environment.**
4. **Apply the algorithms such as clock synchronization, election, and mutual exclusion in distributed applications.**
5. **Identify the challenges in developing distributed applications.**

Q. No.		Max. Marks	CO-BL
1 (A)	How do you organize the Distributed Systems as middleware? Describe with the help of a proper layer diagram.	10	1-3
1 (B)	What do you understand about transparency in Distributed Systems? What are the different forms of transparency in a Distributed System? Describe each with a proper example.	10	1-2
2 (A)	Construct with neat diagrams and give examples of different forms of communication, such as persistent asynchronous, persistent synchronous, transient asynchronous, receipt-based transient synchronous, delivery-based transient synchronous, and response-based transient synchronous communication.	10	2-3
2 (B)	What are the three types of problems in IPC due to different forms of System failures, explain with the help of a neat diagram.	10	2-2
3 (A)	What kind of consistency would you use to implement an electronic stock market? Explain your answer.	10	3-3

	OR Describe a simple implementation of read-your-writes consistency for displaying Web pages that have just been updated.		
3 (B)	In this problem, you are to compare reading a file using a single-threaded file server and a multithreaded server. It takes 15 msec to get a request for work, dispatch it, and do the rest of the necessary processing, assuming that the data needed are in a cache in main memory. If a disk operation is needed, as is the case one-third of the time, an additional 75 msec is required, during which time the thread sleeps. How many requests/sec can the server handle if it is single threaded? If it is multithreaded?	10	3-4
4 (A)	Suppose there are three processes A, B, and C. All clock runs at the same rate but initially, A's clock reads 10, B's clock reads 0 and C's reads 5. At time 10 by A's clock, A sends a message to B, this message takes 4 units of time to reach B. B then waits one unit of time and then sends a message onto C which takes 2 units of time to reach C. Assuming that the system implements Lamport's timestamps draw a picture illustrating the timestamps for the message and explain how the timestamps are obtained.	10	4-5
4 (B)	Why do we need load balancing? What are the different goals of load balancing? What problem does the load balancer solve?	10	4-2
5 (A)	Consider the behavior of two machines in a distributed system. Both have clocks that are supposed to tick 1000 times per millisecond. One of them actually does, but the other ticks only 990 times per millisecond. If UTC updates come in once a minute, what is the maximum clock skew that will occur?	10	5-4
5 (B)	Consider a chain of processes P1, P2, ...,Pn implementing a multitiered client-server architecture. Process Pi is the client of process Pi+1, and Pi will return a reply to Pi −1 only after receiving a reply from Pi +1. What are the main problems with this organization when taking a look at the request-reply performance at process P1?	10	5-5